WASP-11b

R-1618

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_130123` · created 2026-08-02T04:09:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456315.68268 +/- 0.00057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.017
Transit depth (Rp/Rs)^2	1.14 +/- 0.37 [%]
Orbital Inclination (inc)	89.97 +/- 0.77
Airmass coefficient 1 (a1)	1.005 +/- 0.014
Airmass coefficient 2 (a2)	-0.0058 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#1 - [455, 335]
Optimal Aperture	3.61
Optimal Annulus	10.12
Transit Duration (day)	0.1037 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.017 vs 0.131 (deviation 1.13σ)
Duration fit vs published	0.1037 d vs 0.1075 d (deviation 1.21σ)
Mid-time O–C	-1.5 min
Depth SNR	5.0
Residual scatter	1.35 %

Light curve

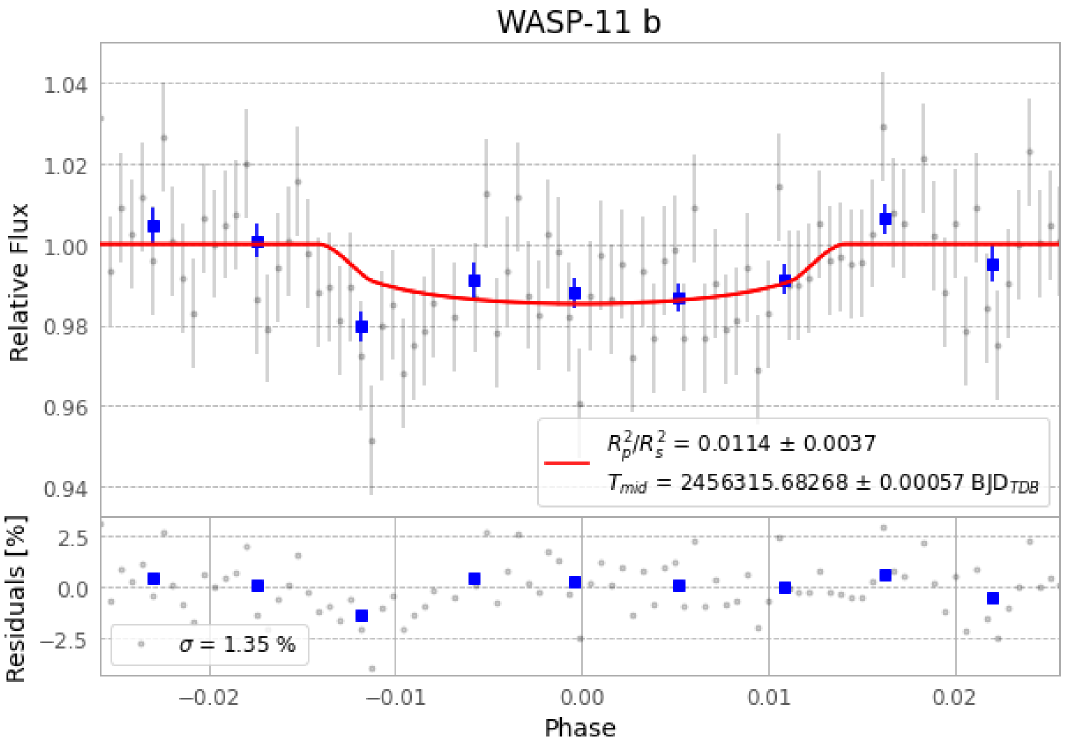

Figure 1 — FinalLightCurve_WASP-11 b_2013-01-22.png (SHA-256 d1fd547c0b6ff73c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 292], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ec78cfa913b4c0fc…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

93 FITS frames (61 MB), `HATP-10130123020012.FITS` → `HATP-10130123063612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-130123.log (f5a63a8dc949…), FinalLightCurve_WASP-11 b_2013-01-22.png (d1fd547c0b6f…), FinalLightCurve_WASP-11 b_2013-01-22.csv (b88a35bb026e…)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

